Fundamental Theorem, Part 1

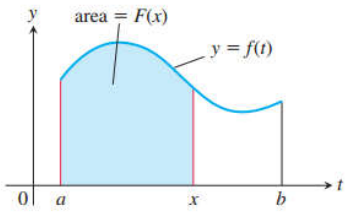


FIGURE 5.19 The function $F(x)$ defined by Equation (1) gives the area under the graph of f from a to x when f is nonnegative and $x > a$.

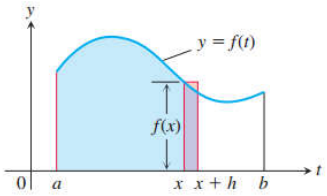


FIGURE 5.20 In Equation (1), $F(x)$ is the area to the left of x. Also, $F(x + h)$ is the area to the left of $x + h$. The difference quotient $[F(x + h) - F(x)]/h$ is then approximately equal to $f(x)$, the height of the rectangle shown here.

THEOREM 4—The Fundamental Theorem of Calculus, Part 1
If f is continuous on $[a, b]$, then $F(x) = \int_a^x f(t)\,dt$ is continuous on $[a, b]$ and differentiable on (a, b) and its derivative is $f(x)$:

$$F'(x) = \frac{d}{dx}\int_a^x f(t)\,dt = f(x). \tag{2}$$

Fundamental Theorem, Part 2 (The Evaluation Theorem)

We now come to the second part of the Fundamental Theorem of Calculus. This part describes how to evaluate definite integrals without having to calculate limits of Riemann sums. Instead we find and evaluate an antiderivative at the upper and lower limits of integration.

THEOREM 4 (Continued)—The Fundamental Theorem of Calculus, Part 2
If f is continuous over $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

EXAMPLE 5 Evaluate the integral $\int_1^3 e^x\,dx$.

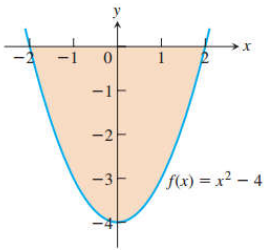
Total Area

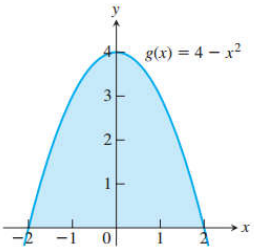
Area is always a nonnegative quantity. The Riemann sum approximations contain terms such as $f(c_k)\,\Delta x_k$ that give the area of a rectangle when $f(c_k)$ is positive. When $f(c_k)$ is negative, then the product $f(c_k)\,\Delta x_k$ is the negative of the rectangle's area. When we add up such terms for a negative function, we get the negative of the area between the curve and the x-axis. If we then take the absolute value, we obtain the correct positive area.

EXAMPLE 6 Figure 5.21 shows the graph of $f(x) = x^2 - 4$ and its mirror image $g(x) = 4 - x^2$ reflected across the x-axis. For each function, compute

(a) the definite integral over the interval $[-2, 2]$, and

(b) the area between the graph and the x-axis over $[-2, 2]$.





EXAMPLE 7 Figure 5.22 shows the graph of the function $f(x) = \sin x$ between $x = 0$ and $x = 2\pi$. Compute

(a) the definite integral of $f(x)$ over $[0, 2\pi]$,

(b) the area between the graph of $f(x)$ and the x-axis over $[0, 2\pi]$.

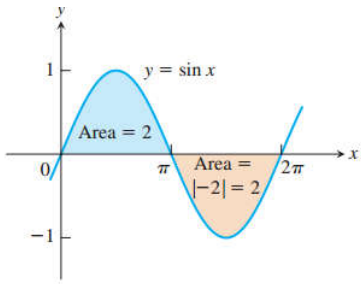


FIGURE 5.22 The total area between $y = \sin x$ and the x-axis for $0 \le x \le 2\pi$ is the sum of the absolute values of two integrals (Example 7).

EXAMPLE 9 What is wrong with the following calculation?

$$\int_{-1}^{3} \frac{1}{x^2}\,dx = \frac{x^{-1}}{-1}\Bigg]_{-1}^{3} = -\frac{1}{3} - 1 = -\frac{4}{3}$$

Summary:
To find the area between the graph of $y = f(x)$ and the x-axis over the interval $[a, b]$:

1. Subdivide $[a, b]$ at the zeros of f.
2. Integrate f over each subinterval.
3. Add the absolute values of the integrals.

EXAMPLE 8 Find the area of the region between the x-axis and the graph of $f(x) = x^3 - x^2 - 2x, -1 \leq x \leq 2$.

EXERCISES 5.4

Evaluating Integrals

Evaluate the integrals in Exercises 1–28.

1. $\int_0^2 x(x-3)\,dx$
2. $\int_{-1}^1 (x^2-2x+3)\,dx$
3. $\int_{-2}^2 \frac{3}{(x+3)^4}\,dx$
4. $\int_{-1}^1 x^{299}\,dx$
5. $\int_1^4 \left(3x^2-\frac{x^3}{4}\right)dx$
6. $\int_{-2}^3 (x^3-2x+3)\,dx$
7. $\int_0^1 \left(x^2+\sqrt{x}\right)dx$
8. $\int_1^{32} x^{-6/5}\,dx$
9. $\int_0^{\pi/3} 2\sec^2 x\,dx$
10. $\int_0^{\pi} (1+\cos x)\,dx$
11. $\int_{\pi/4}^{3\pi/4} \csc\theta\cot\theta\,d\theta$
12. $\int_0^{\pi/3} 4\frac{\sin u}{\cos^2 u}\,du$
13. $\int_{\pi/2}^0 \frac{1+\cos 2t}{2}\,dt$
14. $\int_{-\pi/3}^{\pi/3} \sin^2 t\,dt$
15. $\int_0^{\pi/4} \tan^2 x\,dx$
16. $\int_0^{\pi/6} (\sec x+\tan x)^2\,dx$
31. $\int_2^5 \frac{x\,dx}{\sqrt{1+x^2}}$
32. $\int_0^{\pi/3} \sin^2 x\cos x\,dx$

Derivatives of Integrals

Find the derivatives in Exercises 33–38.

a. by evaluating the integral and differentiating the result.

b. by differentiating the integral directly.

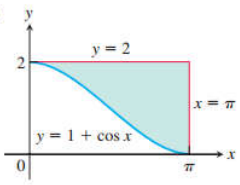
33. $\frac{d}{dx}\int_0^{\sqrt{x}} \cos t\,dt$
34. $\frac{d}{dx}\int_1^{\sin x} 3t^2\,dt$
35. $\frac{d}{dt}\int_0^{t^4} \sqrt{u}\,du$
36. $\frac{d}{d\theta}\int_0^{\tan\theta} \sec^2 y\,dy$
37. $\frac{d}{dx}\int_0^{x^3} t^{-2/3}\,dt$
38. $\frac{d}{dt}\int_0^{\sqrt{t}} \left(x^4+\frac{3}{\sqrt{1-x^2}}\right)dx$

Find dy/dx in Exercises 39–46.

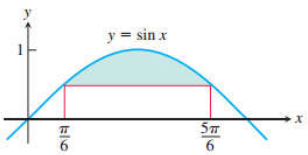
39. $y=\int_0^x \sqrt{1+t^2}\,dt$
40. $y=\int_1^x \frac{1}{t}dt,\quad x>0$
41. $y=\int_{\sqrt{x}}^0 \sin(t^2)\,dt$
42. $y=x\int_2^{x^2} \sin(t^3)\,dt$
43. $y=\int_{-1}^x \frac{t^2}{t^2+4}dt-\int_3^x \frac{t^2}{t^2+4}dt$

Find the areas of the shaded regions in Exercises 51–54.

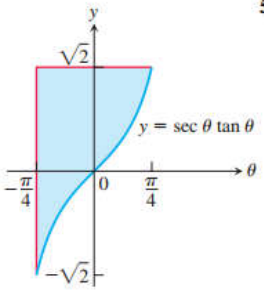
51.



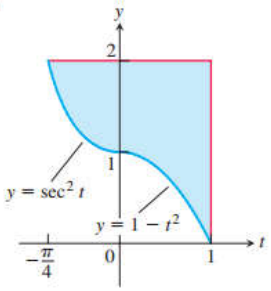
52.



53.



54.



64. Revenue from marginal revenue Suppose that a company's marginal revenue from the manufacture and sale of eggbeaters is

$$\frac{dr}{dx} = 2 - 2/(x + 1)^2,$$

where r is measured in thousands of dollars and x in thousands of units. How much money should the company expect from a production run of $x = 3$ thousand eggbeaters? To find out, integrate the marginal revenue from $x = 0$ to $x = 3$.

65. The temperature T(°F) of a room at time t minutes is given by

$$T = 85 - 3\sqrt{25 - t} \quad \text{for} \quad 0 \le t \le 25.$$

a. Find the room's temperature when $t = 0$, $t = 16$, and $t = 25$.

b. Find the room's average temperature for $0 \le t \le 25$.

66. The height H(ft) of a palm tree after growing for t years is given by

$$H = \sqrt{t + 1} + 5t^{1/3} \quad \text{for} \quad 0 \le t \le 8.$$

a. Find the tree's height when $t = 0$, $t = 4$, and $t = 8$.

b. Find the tree's average height for $0 \le t \le 8$.

67. Suppose that $\int_1^x f(t)\,dt = x^2 - 2x + 1$. Find $f(x)$.

68. Find $f(4)$ if $\int_0^x f(t)\,dt = x \cos \pi x$.

69. Find the linearization of

$$f(x) = 2 - \int_2^{x+1} \frac{9}{1 + t}\,dt$$

at $x = 1$.

7–18 Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of the function.

7. $g(x) = \int_1^x \frac{1}{t^3 + 1}\, dt$

8. $g(x) = \int_3^x e^{t^2 - t}\, dt$

9. $g(y) = \int_2^y t^2 \sin t\, dt$

10. $g(r) = \int_0^r \sqrt{x^2 + 4}\, dx$

11. $F(x) = \int_x^\pi \sqrt{1 + \sec t}\, dt$

$\left[\textit{Hint: } \int_x^\pi \sqrt{1 + \sec t}\, dt = -\int_\pi^x \sqrt{1 + \sec t}\, dt \right]$

12. $G(x) = \int_x^1 \cos \sqrt{t}\, dt$

13. $h(x) = \int_2^{1/x} \arctan t\, dt$

14. $h(x) = \int_0^{x^2} \sqrt{1 + r^3}\, dr$

19–42 Evaluate the integral.

19. $\int_{-1}^{2} (x^3 - 2x)\,dx$

20. $\int_{-2}^{5} 6\,dx$

21. $\int_{1}^{4} (5 - 2t + 3t^2)\,dt$

22. $\int_{0}^{1} \left(1 + \frac{1}{2}u^4 - \frac{2}{5}u^9\right) du$

23. $\int_{0}^{1} x^{4/5}\,dx$

24. $\int_{1}^{8} \sqrt[3]{x}\,dx$

25. $\int_{1}^{2} \frac{3}{t^4}\,dt$

26. $\int_{\pi}^{2\pi} \cos\theta\,d\theta$

27. $\int_{0}^{2} x(2 + x^5)\,dx$

28. $\int_{0}^{1} \left(3 + x\sqrt{x}\right) dx$

29. $\int_{1}^{9} \frac{x - 1}{\sqrt{x}}\,dx$

30. $\int_{0}^{2} (y - 1)(2y + 1)\,dy$

31. $\int_{0}^{\pi/4} \sec^2 t\,dt$

32. $\int_{0}^{\pi/4} \sec\theta \tan\theta\,d\theta$

5.5 Indefinite Integrals and the Substitution Method

The Fundamental Theorem of Calculus says that a definite integral of a continuous function can be computed directly if we can find an antiderivative of the function. In Section 4.8 we defined the **indefinite integral** of the function f with respect to x as the set of *all* antiderivatives of f, symbolized by $\int f(x)\,dx$. Since any two antiderivatives of f differ by a constant, the indefinite integral $\int$ notation means that for any antiderivative F of f,

$$\int f(x)\,dx = F(x) + C, \qquad \text{where } C \text{ is any arbitrary constant.}$$

For example, we can write

$$\int x^2\,dx = \frac{x^3}{3} + C \qquad \text{because} \qquad \frac{d}{dx}\left(\frac{x^3}{3} + C\right) = x^2$$

I TABLE OF INDEFINITE INTEGRALS

$$\int cf(x)\,dx = c\int f(x)\,dx \qquad \int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$$

$$\int k\,dx = kx + C$$

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1) \qquad \int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C \qquad \int a^x\,dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C$$

$$\int \sec^2 x\,dx = \tan x + C \qquad \int \csc^2 x\,dx = -\cot x + C$$

$$\int \sec x \tan x\,dx = \sec x + C \qquad \int \csc x \cot x\,dx = -\csc x + C$$

$$\int \frac{1}{x^2+1}\,dx = \tan^{-1}x + C \qquad \int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1}x + C$$

$$\int \sinh x\,dx = \cosh x + C \qquad \int \cosh x\,dx = \sinh x + C$$

Substitution: Running the Chain Rule Backwards

If u is a differentiable function of x and n is any number different from -1, the Chain Rule tells us that

$$\frac{d}{dx}\left(\frac{u^{n+1}}{n+1}\right) = u^n \frac{du}{dx}.$$

From another point of view, this same equation says that $u^{n+1}/(n+1)$ is one of the anti-derivatives of the function $u^n(du/dx)$. Therefore,

$$\int u^n \frac{du}{dx}\,dx = \frac{u^{n+1}}{n+1} + C. \tag{1}$$

The integral in Equation (1) is equal to the simpler integral

$$\int u^n\,du = \frac{u^{n+1}}{n+1} + C,$$

which suggests that the simpler expression du can be substituted for $(du/dx)\,dx$ when computing an integral. Leibniz, one of the founders of calculus, had the insight that indeed this substitution could be done, leading to the *substitution method* for computing integrals. As with differentials, when computing integrals we have

$$du = \frac{du}{dx}\,dx.$$

EXAMPLE 1 Find the integral $\int (x^3 + x)^5(3x^2 + 1)\,dx$.

EXAMPLE 2 Find $\int \sqrt{2x + 1}\,dx$.

THEOREM 6—The Substitution Rule

If $u = g(x)$ is a differentiable function whose range is an interval I, and f is continuous on I, then

$$\int f(g(x)) \cdot g'(x)\, dx = \int f(u)\, du.$$

The Substitution Method to evaluate $\int f(g(x))g'(x)\, dx$

1. Substitute $u = g(x)$ and $du = (du/dx)\, dx = g'(x)\, dx$ to obtain $\int f(u)\, du$.
2. Integrate with respect to u.
3. Replace u by $g(x)$.

EXAMPLE 3 Find $\int \sec^2(5x + 1) \cdot 5\, dx$

EXAMPLE 4 Find $\int \cos(7\theta + 3)\, d\theta$.

EXAMPLE 3 Find $\int \frac{x}{\sqrt{1 - 4x^2}}\, dx$.

EXAMPLE 1 Find $\int x^3 \cos(x^4 + 2)\, dx$.

EXAMPLE 4 Calculate $\int e^{5x}\, dx$.

EXAMPLE 5 Find $\int \sqrt{1 + x^2}\, x^5\, dx$.

EXAMPLE 6 Calculate $\int \tan x\, dx$.

EXERCISES 5.5

Evaluating Indefinite Integrals

Evaluate the indefinite integrals in Exercises 1–16 by using the given substitutions to reduce the integrals to standard form.

1. $\int 2(2x + 4)^5\,dx, \quad u = 2x + 4$

2. $\int 7\sqrt{7x - 1}\,dx, \quad u = 7x - 1$

3. $\int 2x(x^2 + 5)^{-4}\,dx, \quad u = x^2 + 5$

4. $\int \frac{4x^3}{(x^4 + 1)^2}\,dx, \quad u = x^4 + 1$

5. $\int (3x + 2)(3x^2 + 4x)^4\,dx, \quad u = 3x^2 + 4x$

6. $\int \frac{(1 + \sqrt{x})^{1/3}}{\sqrt{x}}\,dx, \quad u = 1 + \sqrt{x}$

7. $\int \sin 3x\,dx, \quad u = 3x$

8. $\int x \sin(2x^2)\,dx, \quad u = 2x^2$

23. $\int \sec^2(3x + 2)\,dx$

24. $\int \tan^2 x \sec^2 x\,dx$

25. $\int \sin^5 \frac{x}{3} \cos \frac{x}{3}\,dx$

26. $\int \tan^7 \frac{x}{2} \sec^2 \frac{x}{2}\,dx$

27. $\int r^2\left(\frac{r^3}{18} - 1\right)^5\,dr$

28. $\int r^4\left(7 - \frac{r^5}{10}\right)^3\,dr$

29. $\int x^{1/2} \sin(x^{3/2} + 1)\,dx$

30. $\int \csc\left(\frac{v - \pi}{2}\right)\cot\left(\frac{v - \pi}{2}\right)\,dv$

31. $\int \frac{\sin(2t + 1)}{\cos^2(2t + 1)}\,dt$

32. $\int \frac{\sec z \tan z}{\sqrt{\sec z}}\,dz$

33. $\int \frac{1}{t^2}\cos\left(\frac{1}{t} - 1\right)\,dt$

34. $\int \frac{1}{\sqrt{t}}\cos\left(\sqrt{t} + 3\right)\,dt$

35. $\int \frac{1}{\theta^2}\sin\frac{1}{\theta}\cos\frac{1}{\theta}\,d\theta$

36. $\int \frac{\cos\sqrt{\theta}}{\sqrt{\theta}\sin^2\sqrt{\theta}}\,d\theta$

5.6 Definite Integral Substitutions and the Area Between Curves

The Substitution Formula

The following formula shows how the limits of integration change when we apply a substitution to an integral.

THEOREM 7—Substitution in Definite Integrals
If g' is continuous on the interval $[a, b]$ and f is continuous on the range of $g(x) = u$, then

$$\int_a^b f(g(x)) \cdot g'(x)\, dx = \int_{g(a)}^{g(b)} f(u)\, du.$$

EXAMPLE 1 Evaluate $\int_{-1}^{1} 3x^2\sqrt{x^3 + 1}\,dx$.

EXAMPLE 7 Evaluate $\int_{0}^{4} \sqrt{2x + 1}\,dx$ using (6).

EXAMPLE 8 Evaluate $\int_{1}^{2} \frac{dx}{(3 - 5x)^2}$.

EXAMPLE 9 Calculate $\int_{1}^{e} \frac{\ln x}{x}\,dx$.

(a) $\int_{\pi/4}^{\pi/2} \cot\theta \csc^2\theta\,d\theta =$.

(b) $\int_{0}^{\pi/2} \frac{2\sin x \cos x}{(1 + \sin^2 x)^3}\,dx =$.

Definite Integrals of Symmetric Functions

The Substitution Formula in Theorem 7 simplifies the calculation of definite integrals of even and odd functions (Section 1.1) over a symmetric interval $[-a, a]$ (Figure 5.24).

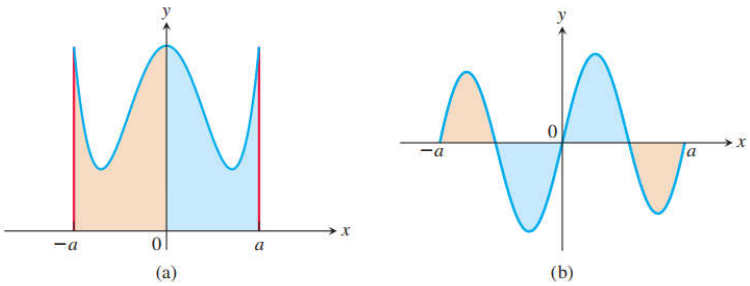


FIGURE 5.24 (a) For f an even function, the integral from $-a$ to a is twice the integral from 0 to a. (b) For f an odd function, the integral from $-a$ to a equals 0.

THEOREM 8 Let f be continuous on the symmetric interval $[-a, a]$.

(a) If f is even, then $\displaystyle\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx.$

(b) If f is odd, then $\displaystyle\int_{-a}^{a} f(x)\,dx = 0.$

EXAMPLE 3 Evaluate $\int_{-2}^{2} (x^4 - 4x^2 + 6)\,dx$.

EXAMPLE 10 Since $f(x) = x^6 + 1$ satisfies $f(-x) = f(x)$, it is even and so

EXAMPLE 11 Since $f(x) = (\tan x)/(1 + x^2 + x^4)$ satisfies $f(-x) = -f(x)$, it is odd and so

5.5 EXERCISES

1–6 Evaluate the integral by making the given substitution.

1. $\int e^{-x}\,dx, \quad u = -x$
2. $\int x^3(2 + x^4)^5\,dx, \quad u = 2 + x^4$
3. $\int x^2\sqrt{x^3 + 1}\,dx, \quad u = x^3 + 1$
4. $\int \frac{dt}{(1 - 6t)^4}, \quad u = 1 - 6t$
5. $\int \cos^3\theta \sin\theta\,d\theta, \quad u = \cos\theta$
6. $\int \frac{\sec^2(1/x)}{x^2}\,dx, \quad u = 1/x$

7–46 Evaluate the indefinite integral.

7. $\int x \sin(x^2)\,dx$
8. $\int x^2(x^3 + 5)^9\,dx$
9. $\int (3x - 2)^{20}\,dx$
10. $\int (3t + 2)^{2.4}\,dt$
11. $\int (x + 1)\sqrt{2x + x^2}\,dx$
12. $\int \frac{x}{(x^2 + 1)^2}\,dx$
13. $\int \frac{dx}{5 - 3x}$
14. $\int e^x \sin(e^x)\,dx$
15. $\int \sin \pi t\,dt$
16. $\int \frac{x}{x^2 + 1}\,dx$
17. $\int \frac{a + bx^2}{\sqrt{3ax + bx^3}}\,dx$
18. $\int \sec 2\theta \tan 2\theta\,d\theta$

19. $\int \frac{(\ln x)^2}{x}\,dx$

20. $\int \frac{dx}{ax+b} \quad (a \neq 0)$

21. $\int \frac{\cos\sqrt{t}}{\sqrt{t}}\,dt$

22. $\int \sqrt{x}\,\sin(1+x^{3/2})\,dx$

23. $\int \cos\theta\,\sin^6\theta\,d\theta$

24. $\int (1+\tan\theta)^5\sec^2\theta\,d\theta$

25. $\int e^x\sqrt{1+e^x}\,dx$

26. $\int e^{\cos t}\sin t\,dt$

27. $\int \frac{z^2}{\sqrt[3]{1+z^3}}\,dz$

28. $\int \frac{\tan^{-1}x}{1+x^2}\,dx$

29. $\int e^{\tan x}\sec^2 x\,dx$

30. $\int \frac{\sin(\ln x)}{x}\,dx$

31. $\int \frac{\cos x}{\sin^2 x}\,dx$

32. $\int \frac{e^x}{e^x+1}\,dx$

33. $\int \sqrt{\cot x}\,\csc^2 x\,dx$

34. $\int \frac{\cos(\pi/x)}{x^2}\,dx$

55. $\int_0^{\pi} \sec^2(t/4)\,dt$

56. $\int_{1/6}^{1/2} \csc\pi t\,\cot\pi t\,dt$

57. $\int_{-\pi/6}^{\pi/6} \tan^3\theta\,d\theta$

58. $\int_0^1 xe^{-x^2}\,dx$

59. $\int_1^2 \frac{e^{1/x}}{x^2}\,dx$

60. $\int_{-\pi/2}^{\pi/2} \frac{x^2\sin x}{1+x^6}\,dx$

61. $\int_0^{13} \frac{dx}{\sqrt[3]{(1+2x)^2}}$

62. $\int_0^{\pi/2} \cos x\,\sin(\sin x)\,dx$

63. $\int_0^a x\sqrt{x^2+a^2}\,dx \quad (a>0)$

64. $\int_0^a x\sqrt{a^2-x^2}\,dx$

65. $\int_1^2 x\sqrt{x-1}\,dx$

66. $\int_0^4 \frac{x}{\sqrt{1+2x}}\,dx$

67. $\int_e^{e^4} \frac{dx}{x\sqrt{\ln x}}$

68. $\int_0^{1/2} \frac{\sin^{-1}x}{\sqrt{1-x^2}}\,dx$

69. $\int_0^1 \frac{e^z+1}{e^z+z}\,dz$

70. $\int_0^{T/2} \sin(2\pi t/T-\alpha)\,dt$

Areas Between Curves